

⇒ Rocket Traj. Sim. Exercise 3 Notes

Starting from the previous exercise, apply altitude-dependent gravity, & solve the system of ODEs using midpoint method & 4th order Runge-Kutta method

> Formulate gravitational acceleration as a function of height from the surface:

Force due to gravity applied on mass m given by

$$\vec{F}_g = \frac{GM_\oplus m}{\|\vec{r}\|^2} \hat{r} = \frac{GM_\oplus m}{(R_\oplus + h)^2} \hat{r}$$

where

$\vec{r}$ = position vector measured from center of Earth = $r_x \hat{i} + r_y \hat{j} + r_z \hat{k}$

G = universal gravitational constant = $6.6743 \times 10^{-11} \text{ [Nm}^2/\text{kg}^2]$

$M_\oplus$ = mass of Earth = $5.9722 \times 10^{24} \text{ [kg]}$

$R_\oplus$ = mean radius of Earth = $6.371 \times 10^6 \text{ [m]}$

h = height above surface of Earth [m]

Acceleration due to gravity is given by

$$\vec{a}_g = \frac{GM_\oplus}{\|\vec{r}\|^2} \hat{r}$$

If 2D, is given by

$$\vec{a}_g = \frac{GM_\oplus}{\|\vec{r}\|^2} \frac{r_x \hat{i}}{\|\vec{r}\|} + \frac{GM_\oplus}{\|\vec{r}\|^2} \frac{r_y \hat{j}}{\|\vec{r}\|}$$

If 1D, is given by

$$\vec{a}_g = \frac{GM_\oplus}{\|\vec{r}\|^2} \frac{r_x \hat{i}}{\|\vec{r}\|}$$

where

$\|\vec{r}\| = \sqrt{r_x^2} = r_x$, thus $\frac{r_x}{\|\vec{r}\|} = \frac{r_x}{r_x} = 1$ & eqn reduces to

$$a_g = \frac{GM_\oplus}{\|\vec{r}\|^2} = \frac{GM_\oplus}{(R_\oplus + h)^2}$$

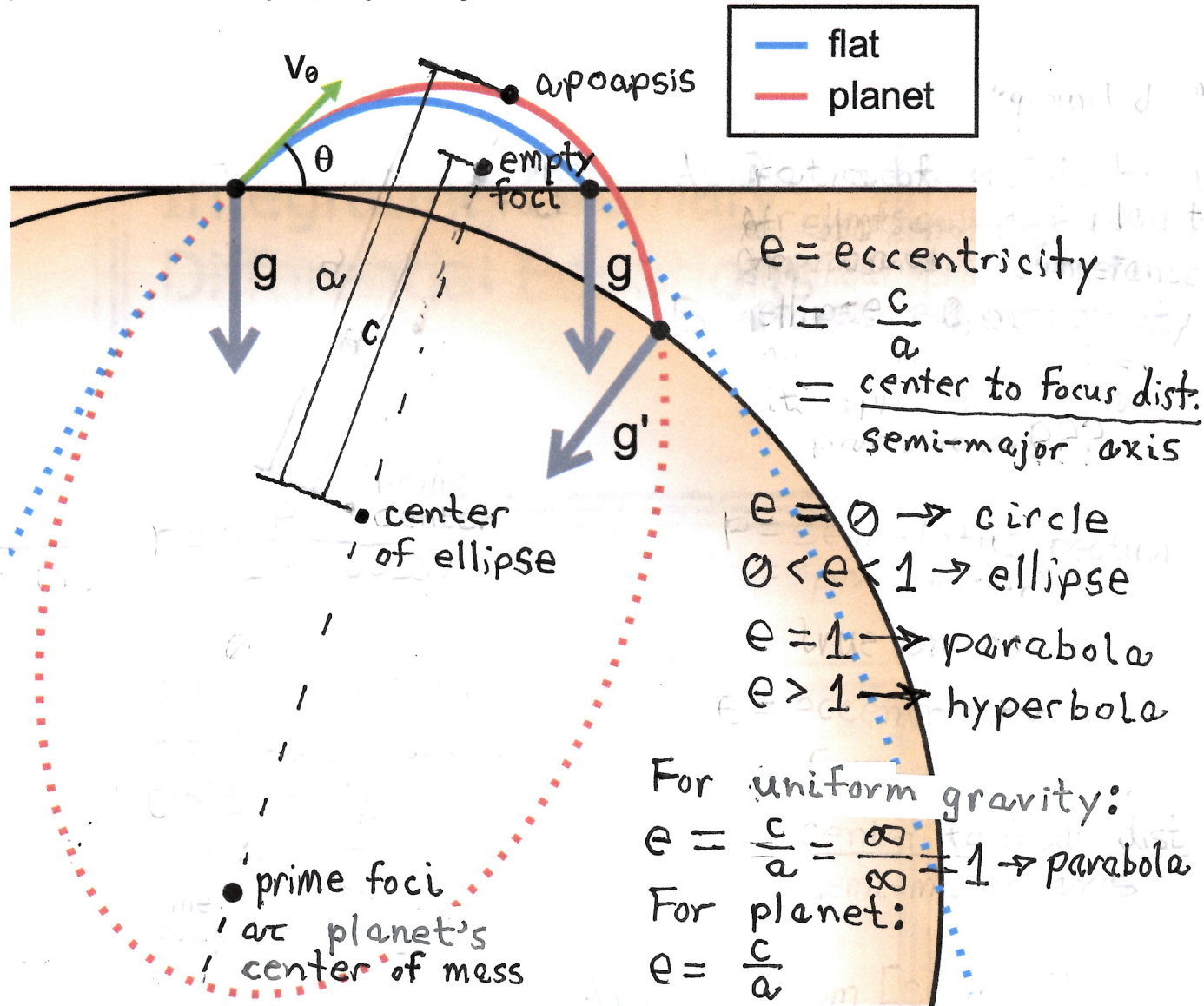
if $h = 0$

$$a_g = \frac{GM_\oplus}{R_\oplus^2} \approx 9.81 \text{ [m/s}^2] = g_0 = \text{standard gravity ;)}$$

Exercise 3 Notes

Visualization of Projectile Motion on a Planetary Scale

Plot of two ballistic trajectories with identical initial velocity, one with a uniform gravity field (blue) and one in the central potential field of a planet (orange). The trajectories are one parabola and one ellipse, respectively.



- Assumptions which yield parabolic trajectory:
 - Earth's gravitational field points straight down relative to launch location (i.e. prime foci at infinity)
- Assumptions which yield elliptical trajectory:
 - Acceleration vector points toward center of Earth (i.e. prime foci NOT at infinity)
- Assumptions which affect impact point:
 - shape of surface (flat, spherical, or oblate spheroid)

(Visualization ref: 'Ballistic-trajectories-planet.svg' by Geek3 on Wikimedia Commons)

Integrating Ordinary Differential Equations

Integrating ODEs

- Tangential, Midpoint and Runge-Kutta Methods
- Simple 1st-Order Example
- Second Order Integrations and Oscillators

Diff Eq Segue

- Similar methods to those of numerical integration carry over into ordinary differential equations (ODEs)
- A great many physical systems can be modeled with these equations
 - orbits
 - ballistics
 - analog circuits
 - springs, dampers
 - pendula

Simple example

- Derivative is just the function itself

$$\frac{dx}{dt} = x$$

$$dx = x dt$$

$$\int \frac{dx}{x} = \int dt \rightarrow$$

$$\ln(x) = t + c$$

- One such function satisfies this:

$$x = Ae^t$$

(A = constant)

$$e^{\ln(x)} = e^{(t+c)}$$

$$x = Ae^t$$

Tangential (Euler) Integration (RK1)

- Consider

$$\frac{dx}{dt} = f(t, x)$$

Basically, just inverting definition of the derivative.

$$\left. \frac{dx}{dt} \right|_{t_0} \approx \frac{x(t_0 + \Delta t) - x(t_0)}{\Delta t}$$

- One could integrate the solution:

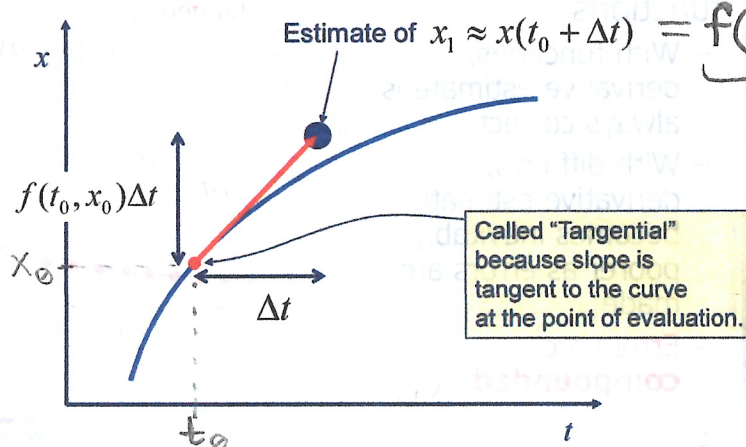
- start with an initial value
- compute derivative
- find next value

$$x(t_0 + \Delta t) \approx x(t_0) + \left(\left. \frac{dx}{dt} \right|_{t_0} \right) \Delta t$$

$$= x(t_0) + f(t_0, x_0) \Delta t$$

Graphically

- Slope** at $t_0, x_0 = f(t_0, x_0)$



$$\frac{dx}{dt} = 0.3x$$

$$\frac{dx}{0.3x} = dt$$

$$\frac{1}{0.3} \int \frac{dx}{x} = t$$

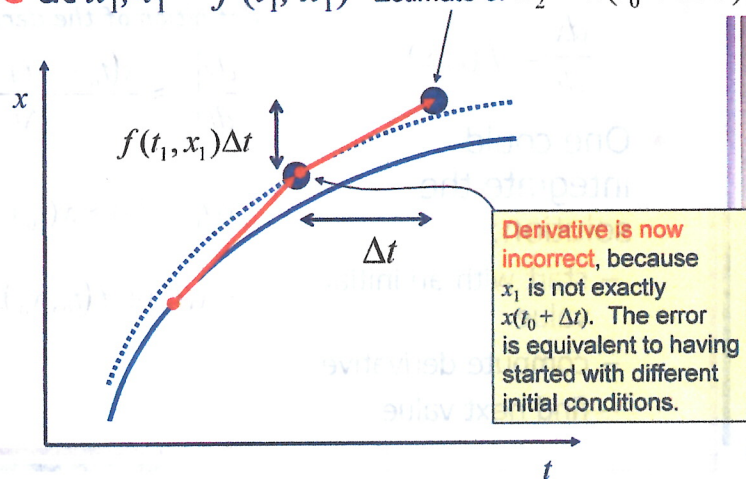
$$\frac{1}{0.3} \ln(x) = t + c$$

$$e^{\ln(x)} = e^{0.3t + 0.3c}$$

$$x = Ae^{0.3t}$$

Graphically

- Slope** at $x_1, t_1 = f(t_1, x_1)$ Estimate of $x_2 \approx x(t_0 + 2\Delta t)$



$$x_n = \frac{dx}{dt} \Delta t + x_{n-1}$$

$$\Delta t = 1, \frac{dx}{dt} = 0.3x$$

Errors Mount

- Errors are worse than in integration of functions

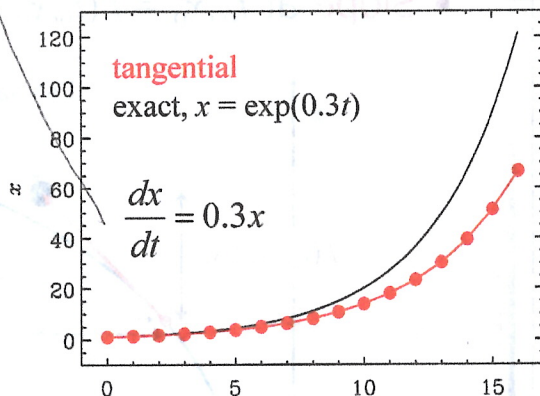
- With functions, derivative estimate is always correct
- With diff eq's, derivative estimate becomes inevitably poorer as errors are made

- Errors are **compounded** (i.e. accumulated)

$$\therefore x_n = 0.3x_{n-1} + x_{n-1}$$

$$= 1.3x_{n-1}$$

$$x_0 = 1$$



OR

$$x_1 = a\Delta t + x_0$$

$$= (a\Delta t + 1)x_0$$

$$x_2 = (a\Delta t + 1)x_1$$

$$= (a\Delta t + 1)^2 x_0$$

$$\vdots$$

$$x_n = (a\Delta t + 1)^n x_0$$

$$= (a\Delta t/n + 1)^n x_0$$

$$f(x, t) = \frac{dx}{dt} \approx \frac{x(t + \Delta t) - x(t)}{\Delta t}$$

$$x_1 = x(t + \Delta t) = \left(\frac{dx(t)}{dt} \right) \Delta t + x(t_0)$$

Quick Check—Euler in the Limit

- Consider the Eulerian treatment for a simple example.

$$\frac{dx}{dt} = ax$$

$$x_1 \approx x_0 + ax_0 \Delta t = [1 + a\Delta t]x_0$$

$$x_2 \approx x_1 + ax_1 \Delta t = [1 + a\Delta t]x_1$$

$$= [1 + a\Delta t]^2 x_0$$

In general:

$$x_n \approx [1 + a\Delta t]^n x_0$$

$$x_n = x(n\Delta t)$$

$$\text{Let } T = n\Delta t$$

$$\therefore x(T)_n = [1 + a\Delta t]^n x_0$$

$$= \left[1 + a \frac{T}{n}\right]^n x_0$$

This is the estimate at $t = T$ for n Eulerian steps

Quick Check—Euler in the Limit

- Now let n get large

$$x(T)_n = \left[1 + a \frac{T}{n}\right]^n x_0$$

$$\lim_{n \rightarrow \infty} x(T)_n = \lim_{n \rightarrow \infty} \left[1 + a \frac{T}{n}\right]^n x_0$$

- Consider the compound interest formula for one year, if compounded n times.

$$\text{amount} = (\text{principal}) \left[1 + \frac{\text{APR}}{n}\right]^n$$

$$A = P \left[1 + \frac{r}{n}\right]^n$$

- Newton's limit shows that

$$\lim_{n \rightarrow \infty} P \left[1 + \frac{r}{n}\right]^n = Pe^r$$

$$\text{let } r = aT, P = x_0$$

$$\lim_{n \rightarrow \infty} x(T)_n = \lim_{n \rightarrow \infty} \left[1 + a \frac{T}{n}\right]^n x_0 = x_0 e^{aT}$$

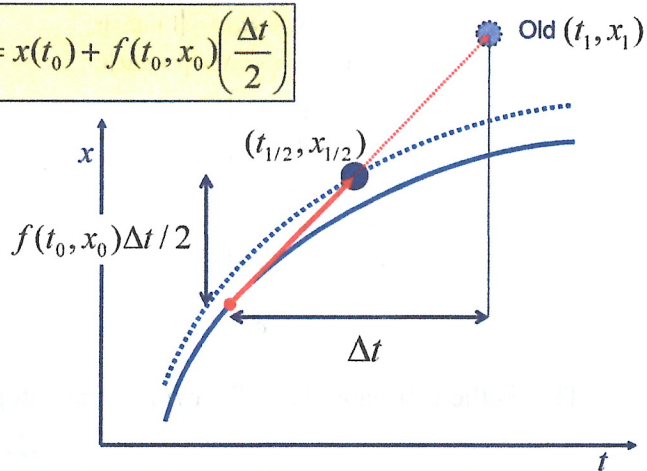


$$e^{at} = \lim_{n \rightarrow \infty} \left(1 + \frac{aT}{n}\right)^n, \quad \Delta t = \frac{T}{n}$$

Midpoint Method (RK2)

- Find $x_{1/2}$ by using slope at x_0 but only moving half a time-step

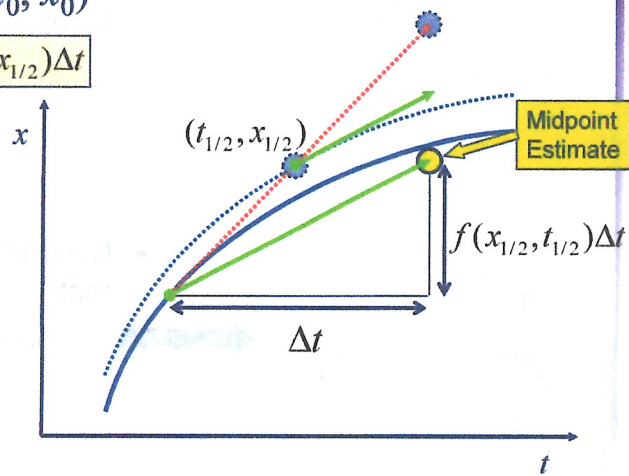
$$x_{1/2} = x(t_0) + f(t_0, x_0) \left(\frac{\Delta t}{2} \right)$$



Midpoint

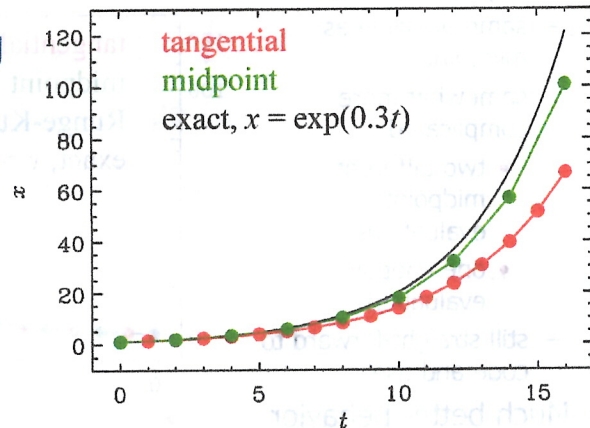
- Use slope at $(t_{1/2}, x_{1/2})$ to propagate full step from (t_0, x_0)

$$x_1 = x(t_0) + f(t_{1/2}, x_{1/2}) \Delta t$$



Midpoint Formula

- Mitigates compounding errors significantly
- Allows for curvature during timestep



Note: 16 calls to derivative function for each

$$\frac{dx}{dt} = ax, \text{ e.g., } \frac{dx}{dt} = 0.3x \rightarrow x = e^{0.3t}$$

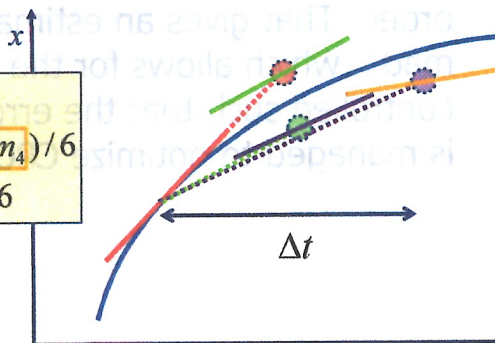
$$x(0) = 1$$

Runge-Kutta (RK4)

- Similar idea to midpoint, but four points
- use slope at start to go to 1st midpoint
- use slope at 1st midpoint from start back to a 2nd midpoint
- use slope at 2nd midpoint to go to endpoint and obtain slope
- Add up slopes thus

$$x_1 = x_0 + \Delta t(m_1 + 2m_2 + 2m_3 + m_4)/6$$

$$x_0 + (k_1 + 2k_2 + 2k_3 + k_4)/6$$



$$m_1 = 0.3x_0$$

$$m_2 = 0.3\left(m_1 \times \frac{\Delta t}{2} + x_0\right)$$

$$m_3 = 0.3\left(m_2 \times \frac{\Delta t}{2} + x_0\right)$$

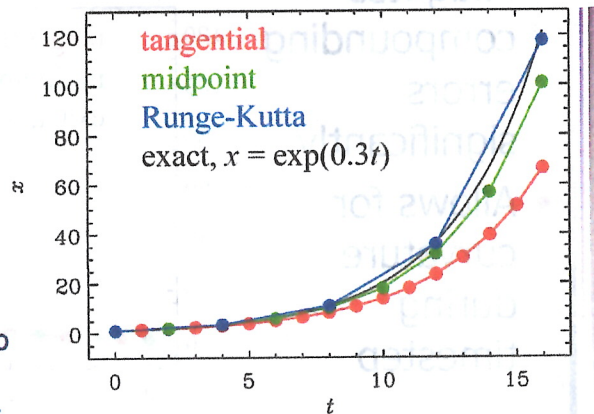
$$m_4 = 0.3(m_3 \times \Delta t + x_0)$$

Runge-Kutta

- Four-point method
 - same principle as midpoint
 - somewhat more complicated
 - two different midpoint evaluations
 - one endpoint evaluation
 - still straight-forward to code and use
- Much better behavior

Exercise: Integrate $\frac{dx}{dt} = 0.3x$ in python:

Exercise_03/ODE-Integration-Examples...
.../compare_ode_int_methods.py



Note: 16 calls to derivative function for each

$$x = 2t^2$$

$$\frac{dx}{dt} = 4t$$

$$\frac{d^2x}{dt^2} = 4$$

Second Order Diff Eq

- Trick facilitates numerical treatment
- Track x and v as independent variables (v is first derivative of x).
- Now set

$$\frac{dx}{dt} = v$$

$$\frac{dv}{dt} = \frac{d^2x}{dt^2} = a$$

$$ma = F$$

$$a = \frac{F}{m}$$



- Now we can use Newton's 2nd Law to integrate all kinds of phenomena

$$V_i \approx \left(\frac{dv}{dt} \right) \Big|_{t_{i-1}} \Delta t + V_{i-1} \Rightarrow x_i = \left(\frac{dx}{dt} \right) \Big|_{t_{i-1}} \Delta t + x_{i-1}$$